

RUBRIC

DS 4002 | Case Study Rubric | University of Virginia

Rubric: Sentiment vs. Star Ratings on Letterboxd

Case Study — NLP Sentiment Analysis | DS 4002, Spring 2024

General Description

In this case study you will apply a pretrained HuggingFace NLP model to a cleaned dataset of Letterboxd movie reviews. You will evaluate how well the model's sentiment predictions align with numeric star ratings, and you will investigate how specific text features influence model accuracy. The deliverable is a reproducible GitHub repository containing your code, data, and a written analysis report.

Preparatory Background

Before beginning, you should be familiar with: Python (pandas, matplotlib, scikit-learn), basic concepts of classification and model evaluation (accuracy, precision, recall, F1), and the idea of pretrained NLP models. No prior NLP experience is required since the model is already trained and ready to use.

Why Am I Doing This?

Sentiment analysis is one of the most deployed NLP techniques in industry. Evaluating a pretrained model on real, messy data, and diagnosing where it fails, is a core data science skill. This case study gives you hands-on experience with the full pipeline: data wrangling, running a transformer model, evaluating performance, engineering diagnostic features, and communicating results clearly.

What Am I Going to Do?

1. Load and Explore the Data Download the cleaned Letterboxd dataset from the repo (letterboxd_top250_reviews_clean.csv). Inspect the dataset: how many reviews are there? What is the distribution of star ratings? Produce at least two EDA visualizations (e.g., rating distribution, review length histogram).

2. Run the Sentiment Model Using the HuggingFace pipeline, run the pretrained RoBERTa sentiment model (cardiffnlp/twitter-roberta-base-sentiment) on the review text column. Store the predicted sentiment label and confidence score for each review.

3. Map Ratings to Sentiment Convert numeric star ratings to sentiment categories: ratings 1–2 = Negative, rating 3 = Neutral, ratings 4–5 = Positive. This becomes your ground truth.

4. Evaluate Overall Performance Compute accuracy and macro F1-score. Produce a confusion matrix. Interpret what the numbers tell you, don't just report them. Why might the numbers be misleading? Discuss class imbalance.

5. Feature Engineering Engineer at least three of the following text features for each review: review length (word count), emoji presence (binary flag), capitalization ratio (proportion of uppercase letters), language detection (English vs. non-English). Compute model accuracy within each subgroup.

6. Interpret and Report Which features are most associated with model errors? Write a 1–2 page analysis interpreting your findings. Address the research question: does textual sentiment reliably align with star ratings, and what features drive disagreement?

Assessment Criteria

Spec Category	Spec Details	Points
Repository Setup	GitHub repo contains README, data folder with CSV, code folder with notebook or .py scripts, and results folder with figures/tables.	15
EDA	At least 2 clearly labeled visualizations of the raw data with brief written interpretation.	10
Model Execution	Sentiment model runs on the full dataset. Predicted labels and confidence scores are stored and reproducible.	15
Evaluation Metrics	Accuracy, macro F1, and confusion matrix are reported and correctly interpreted. Class imbalance is explicitly discussed.	20
Feature Engineering	At least 3 text features are created and subgroup accuracy is computed for each. Features are described clearly.	20

Written Analysis	1–2 page writeup directly addresses the research question. Findings are specific, not generic. Limitations are acknowledged.	20
Reproducibility	A neutral party can clone the repo and reproduce all results by following the README instructions.	10
Total		110

Materials and Resources

- Data: letterboxd_top250_reviews_clean.csv (provided in repo /data folder)
- Starter code: hugging_face_capstone.py (provided in repo /src folder)
- HuggingFace model: cardiffnlp/twitter-roberta-base-sentiment
- Reference: Hutto & Gilbert (2014), VADER: A Parsimonious Rule-based Model for Sentiment Analysis
- Blog explainer: provided in repo /docs folder

Tips for Success

- Run the model on a small sample first (100 rows) to verify it works before running on all 4,755 reviews.
- Class imbalance is the central challenge, be sure to address it head-on in your writeup.
- Feature engineering doesn't need to be complex. A simple binary flag for emoji presence is enough.
- Your written analysis should explain why, not just what.

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